



BHAGWAN MAHAVIR UNIVERSITY
BHAGWAN MAHAVIR COLLEGE OF COMPUTER APPLICATION, SURAT
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[Smart Road]

Abstract :-

- Smart Road with IOT based traffic management, So IOT can help the traffic system to be automated and monitor the traffic status using the internet.
- Automatic street light system is smart and provides a safe night time environment for all road users including pedestrians.
- In earthquake detector, we are monitoring the earthquake, if any vibration/ tilt of the earth's surface takes place; the system will alert us and send the data over IOT page. So to avoid the life loss and economic loss we are developing this device.
- Smart waste management system : In this system, a 24*7 monitoring system is designed for monitoring dumpsters.

Objective :-

- IOT based traffic management solutions for smart cities, If any ambulance will come on a signal then it will show the green path for that ambulance and rest of paths are red.

- When any accidents happens on the road there are buttons used by humans, they push the button & Easy to find the path for emergency condition in ambulance.
- The Traffic violators are captured and send to Police.
- Street lights can switch ON and OFF automatically With the help of IoT.
- LEDs (Light emitting diodes) represents ourlighting system and the amount of light emitted by it is directly related to the amount of light in the environment that is when outside light is less thanthe light given by LEDS is at its full intensity and visa-versa.
- This earthquake early warning system consist of vibration sensor to detect the vibration. It will vibrate when the Earthquake happens.
- If any accident happen the road blocker piezo will active at the time and ring alert alarm.
- Smart waste management system will open door automatically.
- Ultrasonic Sensor is used for measuring the level of waste waste in the dumpster.
- DC Motor powered platform is used for separating wet and dry waste. If either of the container is full then an alert message is sent from the dumpster.
- There are also the recycle type of waste went in Recycle Factory by Escalator.

Tools & Technology :-

- Arduino UNO
- Ultrasonic Sensor
- LED Lights

- Button
- PIR Sensor
- Piezo
- Vibration Sensor
- DC Motor

Conclusion :-

- IOT based traffic management system can help citizens to save their time based on the early-warning messages displayed in the message unit, especially during peak hours. The traffic administration can send priority messages to the citizens; hence, the traffic congestion due to accidents or any such unusual incidents can be avoided.
- Smart Street light Automation system can reduce energy consumption and maintenance costs and also helps to reduce crime activities up to certain limit. This shall reduce heat emissions and replacement costs and carbon dioxide emissions.
- Earthquake system is based on internet of things which make this system more reliable, cost effective and can easily be used by user. This system can be divided into two parts. Firstly the detection of vibration and processed the data accordingly. Secondly, It will vibrate when the Earthquake happens. The main components required in this whole system is Arduino uno which is the heart core of this, vibration sensor.
- Smart waste Management system is very effective in managing waste in any big city. Rather than using conventional periodic collection methods here priority system is used to the city is clean all the time without any

overflowing dumpsters. It has been tested and verified properly to make sure all the different parts work together for a smooth function of the whole system.